Name: _____ Date: _____ Period: _____

3-5: Spot It, Split It

TEKS P.5.J: solve polynomial equations by applying a variety of techniques. Today's technique: factoring a difference of squares.

Warm-up: Mental math

No calculator for the first three columns. Use the middle number, then check with a calculator.

Product	Middle number	Distance from the middle	(middle)² − (distance)²	Calculator check
47 × 53				
38 × 42				
29 × 31				

What do you notice? Why do you think it works?

Part A: Multiply, then find the pattern

Multiply each product. Look at the middle terms.

$(x - 3)(x + 3) =$	$(2x - 5)(2x + 5) =$	$(x + 4)(x - 4) =$

What happened to the middle terms? What do all three answers have in common? Say the pattern in your own words.

Part B: Square roots of the pieces

Fill in the square root of each perfect square. What do you do with the number? With the exponent?

Square	64	$9x^2$	$16m^4$	y^8	$4p^2q^2$	$(1/9)x^2$
Square root						

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Practice: Factor, then check by multiplying back

(B) = Basic, the target for this objective. Factor completely. If an expression is not a difference of squares, say so.

1. $x^2 - 49$ (B)

2. $4x^2 - 25$ (B)

3. $36 - x^2$ (B)

4. $x^6 - 100$ (B)

5. $9y^4 - 64$ (B)

6. $25a^2b^2 - 36$ (B)

7. $\frac{1}{4}x^2 - 9$ (B)

8. $3x^2 - 75$ (B) (GCF first?)

9. $18x^3 - 2x$ (B)

10. $x^4 - 16$ (B) (Factor completely. Look again at each factor.)

11. Alex says $x^2 + 25 = (x + 5)(x + 5)$. Taylor says $x^2 + 25 = (x + 5)(x - 5)$. Multiply both back. Who is right? Can $x^2 + 25$ be factored using real numbers? Explain. (B)

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12. Difference of squares? Yes or No. If yes, factor it; if no, say why not. (B)

Expression	Yes / No	If yes, factor it. If no, why not?
a) $16 - 9x^2$		
b) $x^2 - 24$		
c) $4x^2 - 9x$		
d) $x^3 - 25$		
e) $100m^2 - 1$		

13. Solve by factoring: $9x^2 - 16 = 0$ (B)

14. Use the pattern: 63×57 and 98×102 (B)

15. A square garden has sides of length $2x$ feet. A square pond with sides of 5 feet is built in one corner, so that part is not planted. (B)

(a) Write an expression for the planted area (garden minus pond) and factor it.

(b) The planted area is 75 square feet. Find x.

(c) Cut and slide the planted region into one rectangle (Euclid's picture). Give its dimensions in terms of x, then when $x = 5$.

Stretch stretch: beyond the rubric (for the curious; optional)

16. $(x + 3)^2 - 25$

17. Solve $x^4 - 16 = 0$ (real solutions only)

18. $81x^4 - 16y^4$

Math Time Machine: About 300 BCE, Euclid showed $a^2 - b^2 = (a - b)(a + b)$ with pictures, not symbols. Historians think Babylonian scribes (about 1800 BCE) multiplied using squares, like $47 \times 53 = 50^2 - 3^2$. In 1643 Fermat used differences of squares to factor big numbers: $5959 = 80^2 - 21^2 = 59 \times 101$. Today's internet security (RSA, 1977) depends on factoring a huge number being extremely hard.